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# On the status of *Pseudolytocestus assamensis* Puinyabati *et al* (2013) - a critical review

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**Abstract :** The placement of *Pseudolytocastus assamensis* (Capingentinae: Caryophyllidea: Cestoda) Puinyabati *et al* 2013 has been discussed in the light of works done by Agarwal (1995) Mackiewicz 1985 b and others. Since *Pseudolytocestus* species is reported from Catastomid fishes and are highly endemic & reported from Nearctic region, the report of species like *P. clariae* Gupta (1961), *P. thapari* Gupta and Parmar (1990), *P. fossilisi* Pande *et al* (2000) and *P. assamensis* Puinyabati *et al,* 2013 from Oriental region has become doubtful. The authors have reason to consider *P. assamensis* Puinyabati *et al,* 2013 a possible synonym of *Lytocestus indicus* Moghe (1925). Woodland, 1926.

**Keywords:**

## INTRODUCTION

Puinyabati *et al* (2013)[1] described a new species of a cestode *Pseudolytocestus assamensis* belonging to sub family Capingentinae Hunter (1930)[2]. Wheather the said species is valid or not has been discussed & the authors feel that its placement in Capingentinae is doubtful besides the occurrence of *Pseudolytocestus* from India is also questionable in the light of opinion of Mackiewicz (1981b)[3] who opined that this genus is reported from Nearctic and considering strong endemism of Caryophyllaeids he synonimised *Pseudolytocestus clariae* Gupta (1961) to be a possible synonym of *Capingentoides batrachii* Gupta (1961)[4] if only the so called post ovarian vitelline follicles were ovarian follicles.

Agarwal (1985)[5] too opines "that Gupta (1961)[4] did not study the sections, the nature of hold fast, absence of post ovarian vitellaria and wing like ovary (and not H-shaped) in *P. clariae,* all this lead to *P. clariae* being a possible synonym of *Lytocestus indicus.*"

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## MATERIALAND METHODS

Original literatures have been consulted, and few slides observed.

## OBSERVATION

Description of *Pseudolytocestus assamensis* Puinyabati *et al* (2013)[1] vide Biospectra 8(2): 27-28 is supposed to have following characters: "(body elongated, scolex bluntly round, with short neck, testicular follicles 117-129 in median field anterior to ovary, cirrus sac opening in front of utero-vaginal pore, uterine coil extends little anterior to post end of cirrus sac, ovary H shaped at posterior extremity entirely medullary. Lateral follicles extending into cortex (Rt Lobe - 1.20 - 1.27 × 0.14 - 0.16, Lt. Lobe - 1.00 - 1.25 × 0.11 - 0.19), vitelline follicles cortical partly intruding into medulla in testicular zone & extends from a little posterior to base of neck upto anterior end of ovarian lobes, post ovarian follicles absent, shell gland present")

The description suffers from a number of lacunae. These are :-

(1) The placement of *P. assamensis* in Capingentinae is wrong as no cross section of the worm has

been provided to establish the relationship of inner longitudinal muscle with vitelline follicles, deflection of a few vitelline follicles in testicular zone is possibly on account of pressure exerted while flattening & fixation of the worm else vitellarial follicle intrusion in medulla should have been throughout the length not in testicular zone area alone. The authors of the said species mentions "vitellarial follicles restrict mostly in cortical area, few follicles intrude in testicular zone". This is contrary to the generic diagnosis laid in Yamaguti (1959)[6] who holds that "Vitellaria are situated in peripheral medulla, partly, intruding in cortex".

**Difference between Capingentoides Gupta, 1961 and Pseudolytocestus Hunter, 1929**

| | | *Capingentoides* Gupta, 1961 (*Capingentinae* Hunter, 1930) | ***Pseudolytocestus*** Hunter 1927 (*Capingentinae* Hunter, 1930) |
|---|---|---|---|
| 1 | Scolex | Smooth, oval, truncated anteriorly marked off from rest of body | Little specialised |
| 2 | Neck | long, narrow | short |
| 3 | Testes | Numerous in median field | numerous in broad median field amterior to ovary |
| 4 | cirrus sac | Opens into utero-vaginal canal (atrium) at the beginning of posterior tength of body length | occupies entire thickness of medulla, opens separately on ventral surface or into shallow eversible atrium a little in front of utero-vaginal pore |
| 5 | Ext. Sem-Vesicle | | present |
| 6 | int. Sem-vesicle | Large Conical bell shaped | - |
| 7 | uterine and Vaginal pore | Common | Common |
| 8 | Vitellaria | partly cortical & partly medullary | in Peripheral medulla of testicular zone party intruding into cortex |
| 9 | Ovarian follicles | cortical | H-shaped at posterior extremity almost entirely medullary, its lateral follicles extend into cortex |
| 10 | Isthmus | medullary | medullary |
| 11 | Uterine Coils | never extend anterior to cirrus sac | never extend forward beyond cirrus pouch |
| 12 | Uterine glands | Present | N.M |
| 13 | Receptaculum Seminis | absent | N.M |
| 14 | Post ovarian follicles | Present | No post ovarian follicles |
| 15 | Parasitic in | Clariids | Catostomid fishes |

In *P. differtus* too vagina joins the uterus to form a utero-vaginal duct, opening posterior to the male gonopore (vide fig. below)

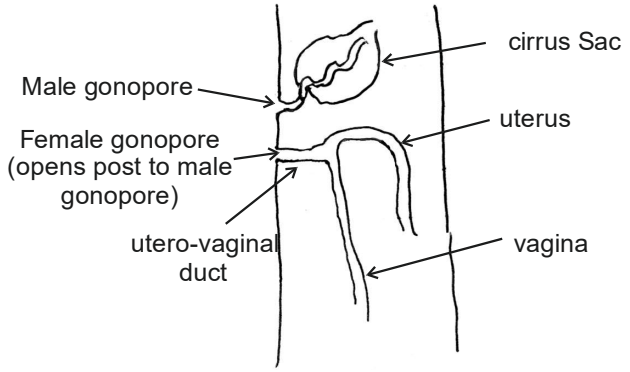



Fig. 1: Male and female gonopore separate

In *P. assamensis* Ovary is said to be is H-shaped & distinctly bilobed at posterior extremity, almost entirely medullary with its lateral follicles extending into cortex (Rt. OV. Lobe - 1.20 - 1.27 × 0.14 - 16 mm) Lt. OV. Lobe 1.00 - 1.25 × 0.11 - 0.19 nevertheless it is cortical in the camera lucida drawing On page 28 Biospectra Vol. 08(2), 2013; isthmus being medullary. If a reference to the camera lucida drawing of *Pseudolytocestus clariae* Gupta, (1961)[4] is made ovary is definitely medullary (ovary H-Shaped distinctly lobed, isthmus more or less at posterior end of ovary. Left wing of ovary narrower in outline, slightly lobed and rounded at ends, 1.12 × 0.4 mm, right wing of ovary broader, slightly lobed & pointed at ends 1.06 × 0.6 mm. Isthmus as well as ovary both are medullary). In generic diagnosis ovary is medullary.

Uterine coils in *P. assamensis* is said to be restricted behind cirrus sac, however in *P. clariae* it extends upto a little posterior to anterior of cirrus sac (In generic diagnosis uterine coils never extend forward beyond cirrus pouch).

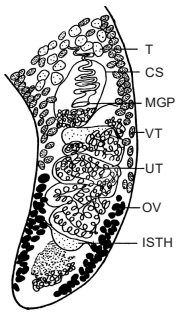



Fig. 2: *Pseudolytocestus assamensis* (Posterior end) Puinyabati *et al,* 2013

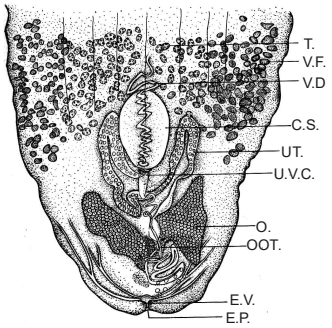



Fig. 3: *Pseudolytocestus clariae* (Posterior end) Gupta, 1961

The occurrence of Pseudolytocestus species from oriented region is perplexing in the light of Mackiewicz 1981b opinion that it is endemic & is restricted to Nearctic.

## CONCLUSION

*Pseudolotocestus assamensis* Puinyabati *et. al.,* 2013 does not belong to Capingentinae because. :-

1. Intrusion of vitelline follicles in testicular zone has been shown, not through out the length of the cestode, possibly it is due to pressure exerted while flattening & fixation. No cross section of the worm has been provided to ascertain the inner longitudinal muscle relationship vis-à-vis vitelline follicles, ovary etc.
2. Ovary has been shown to cortical in camera lucida drawing (This is against the generic diagnosis, ovary is medullary in Pseudolytocestus.
3. The occurrence of *Pseudolytocestus* from Oriental region is doubtful as it is restricted to Nearctic.
4. In the light of the nature of hold fast, absence of post ovarian vitellaria & nature of ovary (cortical) *Pseudolytocestus clariae* Gupta (1961)[4], *P. thapari* Gupta & Parmar (1990)[7], *P. fossilisi* Pande *et al* (2000) and *P. assamensis* Puinyabati *et al* (2013)[1] are a possible synonym of *Lytocestus indicus* Moghe (1925, 31)[9] Woodland, 1926[10] as suggested by Agarwal (1985)[5].

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